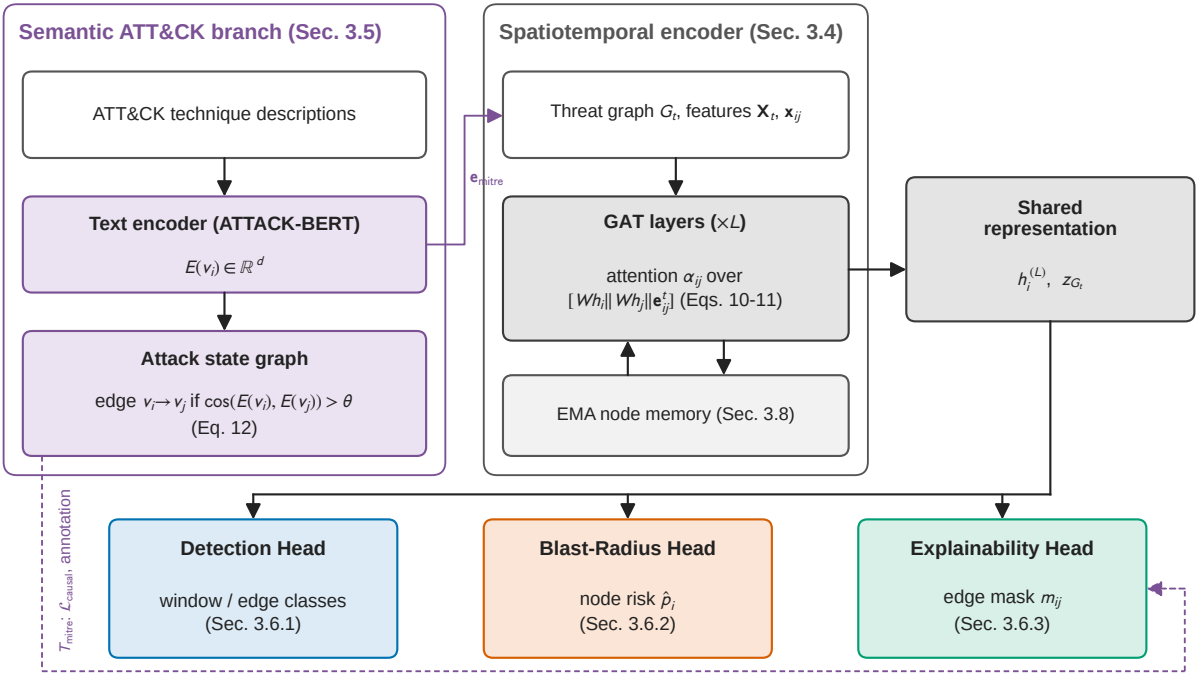
Semantic ATT&CK branch (Sec. 3.5)
ATT&CK technique descriptions
Text encoder (ATTACK-BERT)
$E(v_i) \in \mathbb{R}^d$
Attack state graph
edge $v_i \to v_j$ if $\cos(E(v_i), E(v_j)) > \theta$
(Eq. 12)
$\mathbf{e}_{\text{mitre}}$
Spatiotemporal encoder (Sec. 3.4)
Threat graph G_t, features $\mathbf{X}_t$, $\mathbf{x}_{ij}$
GAT layers ($\times L$)
attention α_{ij} over
$[Wh_i \| Wh_j \| \mathbf{e}^t_{ij}]$ (Eqs. 10-11)
EMA node memory (Sec. 3.8)
Shared representation
$h_i^{(L)}$, z_{G_t}
T_{mitre}, $\mathcal{L}_{\text{causal}}$, annotation
Detection Head
window / edge classes
(Sec. 3.6.1)
Blast-Radius Head
node risk $\hat{\rho}_i$
(Sec. 3.6.2)
Explainability Head
edge mask m_{ij}
(Sec. 3.6.3)